

Internal AI Adoption Proposal

1. Purpose and requirements

This document answers:

- (1) Options for a local AI server on open-source models, accessed over VPN, with hardware configurations;
- (2) The paid-service options and what sharing one account across a team really looks like;
- (3) Other options;
- (4) An implementation and training plan.

Identified AI use cases:

- Testing, validation and automated status summaries.
- Analysis of technical problems.
- Code review, coding-style fixes, adding comments.
- Customer support.
- Recurring market intelligence: UWB, RTLS, market progress and demand by industry.

2. Option 1 - Local AI server with open-source models

The most secure option: all data (source code, customer documents) never leaves company infrastructure. Staff connect over VPN from the office or remotely.

2.1. Architecture

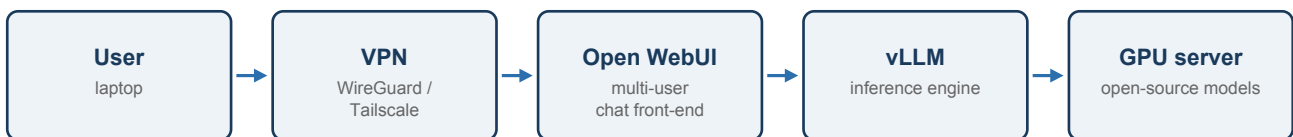


Figure 1: Access architecture for the local AI server over VPN.

- **OS:** Ubuntu Server 24.04 LTS + Docker.
- **Inference engine:** vLLM for multi-user serving (PagedAttention, high throughput); Ollama only suits personal experiments.
- **Front-end:** Open WebUI - open source, multi-user with role-based access, per-user conversation isolation, built-in RAG (Q&A over internal documents), SSO/LDAP support.
- **API gateway (optional):** LiteLLM proxy - virtual keys per person with budgets, one endpoint combining local and cloud models.
- **VPN:** Tailscale (near zero-config, free up to 3 users/100 devices) or self-hosted WireGuard (completely free).

Sources / setup guides:

- <https://docs.openwebui.com/>
- <https://docs.vllm.ai/en/latest/deployment/frameworks/open-webui/>
- <https://docs.litellm.ai/>
- <https://tailscale.com/pricing>
- <https://www.wireguard.com/>

2.2. Notable open-source models

Top open-source LLMs ranked by aggregated benchmarks (benchlm.ai, June 2026):

Rank	Model	Creator	Overall	Context
1	DeepSeek V4 Pro (Max)	DeepSeek	87	1M
2	Kimi K2.6	Moonshot AI	84	256K
3	GLM-5 (Reasoning)	Zhipu AI	81	200K
4	GLM-5.1	Zhipu AI	83	203K
5	DeepSeek V4 Pro (High)	DeepSeek	83	1M
6	Qwen3.5 397B (Reasoning)	Alibaba	77	128K
7	DeepSeek V4 Flash (Max)	DeepSeek	77	1M
8	Qwen3.6-27B	Alibaba	75	262K

The top-ranked models have hundreds of billions of parameters and need large GPU infrastructure; for the server configurations in section 2.3, the smaller variants (e.g. Qwen3.6-27B) or distilled/quantized versions of these models are the practical fit.

Quick coding-capability comparison of the open models (higher is better, source: benchlm.ai):

Model	Blended coding score	LiveCodeBench	SWE-bench Verified
DeepSeek V4 Pro (Max)	89.8	93.5	-
Kimi K2.6	88.7	89.6	-
DeepSeek V4 Pro (High)	88.7	-	-
Qwen3.5 397B (Reasoning)	86.7	-	-
GLM-5.1	84.1	-	77.8
GLM-5 (Reasoning)	-	-	77.8
DeepSeek V4 Flash (Max)	-	91.6	-

Reference:

- <https://benchlm.ai/blog/posts/best-open-source-llm>
- <https://huggingface.co/blog/daya-shankar/open-source-llms>

Evidence comparing coding capability across models (continuously updated leaderboards, covering both open and commercial models):

- <https://www.swebench.com/>
- <https://aider.chat/docs/leaderboards/>
- <https://livecodebench.github.io/leaderboard.html>
- <https://evalplus.github.io/leaderboard.html>

2.3. Reference hardware configurations

Item	Tier A - Entry	Tier B - Mid-range	Tier C - Mac alternative
GPU	1× NVIDIA RTX 5090 32 GB GDDR7	2× RTX 5090 (64 GB total) or 1× RTX PRO 6000 Blackwell 96 GB	Apple M3 Ultra, 80-core GPU, unified memory
CPU	Ryzen 9 7950X / Core i9 (8+ cores)	Threadripper / Xeon W (enough PCIe lanes for 2 GPUs)	-
RAM	64 GB DDR5	128-256 GB DDR5 ECC	256-512 GB unified memory
Storage	2 TB NVMe	4 TB NVMe (models + RAG data)	2-4 TB SSD
PSU / misc	1000 W PSU	1600 W PSU, well-ventilated case, UPS	Very low power draw
Models it can run	≤ 32B Q4 (Qwen3 32B, Gemma 3 27B, gpt-oss-20b)	70B Q4, gpt-oss-120b, quantized Qwen3 235B-A22B	Very large models (DeepSeek R1 671B at ~17-18 tokens/s)

Item	Tier A - Entry	Tier B - Mid-range	Tier C - Mac alternative
Concurrent users	~3-5	~10-20 with vLLM	1-3 (weak under concurrency)
Evidence that the hardware fits the models	An RTX 5090's 32 GB is enough for quantized ~30B models: runpod.io (RTX 5090 LLM guide)	70B Q4 needs ~40 GB VRAM; 2× RTX 5090 or a 96 GB card covers it: runpod.io (RTX 5090 LLM guide) , bizon-tech.com (best GPU for LLM)	The M3 Ultra 512 GB runs DeepSeek R1 671B at ~17-18 tokens/s: medium.com (M3 Ultra LLM inference) , apple.com/mac-studio/specs
Estimated cost	≈ US\$2,500-3,500	≈ US\$7,000-12,000	≈ US\$5,600 (256 GB) - \$9,500 (512 GB)

Basis: a 70B model at FP16 needs ~140 GB of VRAM; 4-bit quantization (Q4_K_M) reduces this to ~40 GB - more than one RTX 5090's 32 GB, hence two GPUs or a 96 GB card. The Mac Studio can hold very large models thanks to unified memory, but its long-prompt (prefill) speed and concurrency are weaker than NVIDIA GPUs.

Reference:

- <https://www.nvidia.com/en-us/geforce/graphics-cards/50-series/rtx-5090/>
- <https://www.nvidia.com/en-us/products/workstations/professional-desktop-gpus/rtx-pro-6000/>
- <https://www.runpod.io/articles/guides/nvidia-rtx-5090>
- <https://bizon-tech.com/blog/best-gpu-llm-training-inference>
- <https://www.apple.com/mac-studio/specs/>
- <https://medium.com/@billynewport/apples-m3-ultra-mac-studio-misses-the-mark-for-llm-inference-f57f1f10a56f>

2.4. Assessment

Strengths	Limitations
Data stays 100% on-premises; no per-seat subscription fees; unlimited usage; full control and customization (internal-document RAG, fine-tuning).	Open 32B-120B models still trail top commercial models on hard tasks; needs an operator (model updates, backups, monitoring); no built-in web search unless integrated; upfront hardware investment.

3. Option 2 - Paid AI services (monthly / yearly)

3.1. The options

Service	Price/user/month	Notes
ChatGPT Business (OpenAI) openai.com/chatgpt-pricing	US\$25 (monthly) / US\$20 (annual)	Minimum 2 seats; dedicated workspace; data not used for training
Claude Team Standard (Anthropic) claude.com/pricing	US\$25 (monthly) / US\$20 (annual)	Minimum 5 seats; SSO, central admin; no training on team data
Claude Team Premium claude.com/pricing	US\$125 (monthly) / US\$100 (annual)	Standard + Claude Code (terminal/IDE coding agent)
Cursor Pro / Teams cursor.com/pricing	US\$20 (Pro) / US\$40 (Teams)	AI coding IDE; each month includes a model-usage allowance matching the plan price
GitHub Copilot Business github.com/copilot/plans	US\$19	In-IDE code completion; complements rather than replaces AI chat



Figure 2: Coding capability of the model behind each service - SWE-bench Verified score (% of real GitHub issues solved), June 2026. Cursor and Copilot run these same models, so coding ability follows the chosen model.

Evidence comparing coding capability of commercial models (Claude, GPT, Gemini...) on independent leaderboards:

- <https://benchlm.ai/benchmarks/sweVerified>
- <https://www.swebench.com/>
- <https://aider.chat/docs/leaderboards/>
- <https://lmarena.ai/leaderboard>
- <https://livecodebench.github.io/leaderboard.html>

3.2. Sharing one account across the team - possible, but limits hit fast

Technically several people can log into one account, but **usage limits are counted per account**, so a shared account runs dry very quickly:

- **Claude (Pro/Max):** a rolling **5-hour session limit** (starting from the first prompt) plus a **weekly limit**; every channel (web, IDE, Claude Code) draws from the same quota. Pro allows only roughly 10-45 prompts per 5-hour session depending on length. Source: support.claude.com - [How do usage and length limits work](#).
- **ChatGPT Plus:** about **160 GPT-5.5 messages every 3 hours**; beyond that it falls back to the mini model. Source: help.openai.com - [GPT-5.5 in ChatGPT](#). The Business plan is virtually unlimited on base models, but is billed per seat. Source: help.openai.com - [ChatGPT Business Models & Limits](#).
- **Cursor:** limited **per month** - the US\$20 Pro plan includes ~US\$20 of model usage each month; when it is used up, it stops (or continues on-demand at extra cost if a spend limit is set). Sources: cursor.com/pricing, cursor.com/blog/june-2025-pricing.
- **Limits depend on context/tokens, not message count:** providers meter quota by tokens processed. Long conversations, large file attachments and big code contexts burn quota many times faster - so the more people share one account, the sooner it hits the ceiling.
- Also, sharing a personal account falls outside the services' terms of use; the official way for a team is a per-seat Business/Team plan.

4. Other options

4.1. Renting cloud GPUs

Rent GPUs by the hour to run the same open-source models from section 2. Advantages:

- No upfront investment - pay only for hours actually used.
- Try several GPU configurations before committing to a hardware purchase.
- Scale up or down flexibly; rent a bigger machine for a few hours when a heavy job comes up.
- Usable immediately while waiting for purchased hardware to arrive.

4.2. Comparison matrix

Criterion	Option 1 - Local server	Option 2 - Paid services	Cloud GPU rental
Data security	Highest - data never leaves the company	Depends on vendor (no-training commitments)	Data passes through the rental provider

Criterion	Option 1 - Local server	Option 2 - Paid services	Cloud GPU rental
AI quality	Good (open 32B-120B models)	Best (latest commercial models)	Same as Option 1 (same open models)
Upfront cost	US\$2,500-12,000 by tier	0	0
Monthly cost	Power + maintenance	US\$20-125/user by plan	Per hour of use
Usage limits	None	Session/weekly/monthly depending on service (3.2)	None (within rented hours)
Web search	Requires extra integration	Built-in	Requires extra integration
Operating effort	Needs an owner	Negligible	Moderate

5. Implementation plan

Timeline	Key activities	Outcome
Week 1, days 1-2	Confirm option and budget; define which data may/may not go to cloud; open accounts; deploy Open WebUI (+ LiteLLM if using APIs) on an interim machine; set up Tailscale/WireGuard VPN	Whole team starts using AI via an internal web app
Week 1, days 3-5	Stand up the open-model server: Ubuntu + Docker + vLLM, pull models (Qwen3 32B, Qwen2.5-Coder); if hardware is not yet available, run on a rented cloud GPU	Open-source models live, reachable over VPN
Week 2	Integrate use cases: AI code review on Git, internal-document RAG, test-summary scripts, weekly UWB/RTLS market-watch agent; record demo videos (section 6); publish the AI usage policy	AI embedded in daily workflows, guides available

If hardware must be ordered, shipping time falls outside these 2 weeks - use rented GPUs meanwhile.

6. Team training

- **Record setup demo videos** (5-10 min screen recordings): installing and connecting the VPN, logging into Open WebUI, connecting the IDE/terminal agent, configuring internal API keys - stored on the internal drive so new staff can set themselves up anytime.
- **Share a curated list of prompt-writing guides** with the team:

Prompt-engineering guides:

- <https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/overview>
- <https://platform.openai.com/docs/guides/prompt-engineering>
- <https://www.promptingguide.ai/>

- A one-page cheat sheet of ready-to-use prompts for company work.

7. Mapping use cases → solutions

Use case	Solution	Model / tool
Testing, validation, status summaries	Scripts calling the internal model API: read test/CI logs, generate a daily summary posted to chat	Qwen3 32B (local) - internal data
Problem analysis	Direct chat; escalate hard problems to a stronger model	Claude/GPT (cloud) or DeepSeek-R1-Distill (local)
Code review, style, comments	Wired into Git: automatic AI review on every merge request; engineers use a terminal/IDE agent	Qwen2.5-Coder (local), Claude Code / Cursor (cloud)
Customer support	A Q&A chatbot over the docs.leapslabs.com documentation using RAG (details in section 9)	Open WebUI Knowledge (local)

Use case	Solution	Model / tool
UWB/RTLS market watch	A script crawls news from a fixed source list; AI only summarizes and classifies; notifications are sent (details in section 8)	Crawl script + local model or API
New application development	Build one small app in a 2-week pilot with AI pair-programming to measure real speed - separate discussion to pick the pilot	Claude Code / Cursor + standard project template

8. UWB/RTLS market intelligence monitoring

We build a script that crawls news from a fixed list of websites; AI handles **summarization and classification**; the system then sends notifications. A 4-step pipeline:

- **Step 1 - Fix the source list:** the websites/RSS feeds in the table below; adjustable over time.
- **Step 2 - News crawl script:** Python ([feedparser](#) for RSS; requests + [BeautifulSoup](#) for pages without RSS), run daily by cron, storing new articles in a database with de-duplication. A no-code alternative is [n8n](#) (self-hostable).
- **Step 3 - AI summarization and classification:** for each new article, the model writes a 3-5 sentence summary and assigns labels (UWB / RTLS / chips and hardware / competitors / market reports / industry applications: automotive, healthcare, logistics...), scores relevance 1-5 and drops articles below the threshold. News is public data, so either the local model (Qwen3 32B) or an API works.
- **Step 4 - Notification:** a weekly digest via email/Slack/Teams; highly relevant articles forwarded the same day.

Suggested source websites:

Source	Main content
FiRa Consortium firaconsortium.org	UWB standards body: certification news, new members, standard use cases
UWB Alliance uwballiance.org	UWB ecosystem, spectrum policy advocacy
omlox omlox.com	Open locating standard for industry (UWB at its core)
RFID Journal rfidjournal.com	RTLS and asset-tracking news, real deployment case studies in logistics, healthcare, manufacturing
everything RF everythingrf.com/news	Newly released RF and UWB chips/modules
NXP Newsroom nxp.com (newsroom)	Trimention UWB chips, automotive and smartphone partnerships
Qorvo Newsroom qorvo.com/newsroom	DW3000 UWB chip line (ex-Decawave) and industry news
Sewio Blog sewio.net/blog	UWB RTLS in factories; positioning-technology comparisons (competitor watch)
MarketsandMarkets / Grand View Research marketsandmarkets.com (UWB report) , grandviewresearch.com (UWB report)	Periodically updated UWB market sizing and forecast reports
Google News RSS news.google.com/rss/search?q=UWB+OR+RTLS	Broad keyword sweep, free, easy to add new keywords (e.g. competitor or industry names)

9. Customer support: a Q&A chatbot over docs.leapslabs.com

Build a chatbot that lets users ask questions directly, with all of its knowledge taken from docs.leapslabs.com. The chatbot works on the **RAG** (Retrieval-Augmented Generation) pattern: it answers only from the documentation content, citing links to the relevant docs pages; when the answer is not in the docs it says 'not covered in the documentation' instead of making something up.

How to build it (pipeline):

- **Step 1 - Ingest the docs:** a script downloads all pages of docs.leapslabs.com (via the sitemap), converts them to markdown and splits them into chunks; re-run automatically every week by cron so updates to the docs are picked up.
- **Step 2 - Index:** create embeddings for each chunk and store them in a vector database. Open WebUI ships with a built-in Knowledge (RAG) feature, so nothing has to be built from scratch.
- **Step 3 - Q&A:** when a user asks a question, the system retrieves the most relevant doc chunks, puts them into context, and the model answers with source links.
- **Step 4 - Two-stage rollout:** (a) internal first: the support team uses the chatbot to draft replies to customers (human approves before sending); (b) once quality is stable, embed a public chat widget on the website/docs so customers can ask directly, with hard questions automatically handed over to a human.
- Log every customer question: it both measures answer quality and shows which parts of the docs are missing and should be added.
- The docs content is public, so either the local model (Qwen3 32B) or an API works; running locally means no per-question cost.

Tools / frameworks for building the RAG chatbot:

- <https://docs.openwebui.com/>
- <https://docs.llamaindex.ai/>
- <https://python.langchain.com/docs/>

10. Risks and mitigations

Risk	Mitigation
Sensitive data leaking to cloud services	Data classification policy; confidential data restricted to local models
Incorrect AI answers (hallucination)	Human review before anything reaches customers; require sources
Costs running out of control	Per-user limits; weekly cost review
Hardware failure / overload	UPS and config backups; GPU monitoring; temporary cloud fallback during maintenance
Low adoption after training	Replayable demo videos; AI review built into the merge-request process

Prices and specifications researched in June 2026 and subject to change; re-confirm at purchase time.